Models of Language and Conversation

Lecture 22: Scaling Up to GPT-3 and Beyond

The Era of Few-Shot Learning

Models of Language and Conversation

Week 7

Week 7

Today's Journey

1. **GPT-2** - 10x scale-up, zero-shot multitask learning
2. **GPT-3** - 100x scale-up, few-shot learning emerges
3. **Scaling Laws** - Why bigger is (predictably) better
4. **RLHF & ChatGPT** - Aligning LLMs with human preferences
5. **Modern Landscape** - Open vs closed models
6. **Hands-on Examples** - Prompting techniques in practice

GPT-2: The Unexpected Leap

Discussion

What happens when we scale up GPT by 10x?

GPT (2018):

- 117M parameters
- 5B tokens training
- BooksCorpus
- Requires fine-tuning

GPT-2 (2019):

- 1.5B parameters (13x larger)
- 40GB text (8x more data)
- WebText dataset
- **No fine-tuning needed!**

Key Claim

"Language models are unsupervised multitask learners"

The WebText Dataset

How GPT-2 was trained:

WebText Creation

1. Scrape all outbound links from Reddit with >=3 karma
2. Filter for quality and diversity
3. Remove Wikipedia (to avoid test set contamination)
4. Result: 40GB of text, 8 million documents

Why Reddit links?

- Community curation (karma = quality signal)
- Diverse topics and writing styles
- Web-scale variety

WebText: Concrete Examples

What kinds of documents were included:

```
Reddit post (3+ karma): "Check out this great article about climate
```

Sample document types in WebText:

Source Type	Example	Why Included
News articles	NYT, BBC	High quality journalism
Educational	Medium posts,	Clear

GPT-2 Model Sizes

Four model sizes released progressively:

Model	Parameters	Layers	Hidden Size
Small	117M	12	768
Medium	345M	24	1024
Large	762M	36	1280

Zero-Shot Task Transfer

The surprising finding: GPT-2 can perform tasks without fine-tuning!

Zero-Shot Prompting Examples

Translation:

```
English: I love machine learning
French:
```

GPT-2 completes: "J'aime l'apprentissage automatique"

Question Answering:

```
Answer the question:
Q: What is the capital of France?
A:
```

Zero-Shot: How It Works

GPT-2 saw similar patterns during pre-training:

```
Example from training data (hypothetical):
```

At inference time:

```
User prompt: "[Article about climate conference]
```

Key insight: Zero-shot works because the model learned task formats implicitly from diverse web text!

GPT-2 Performance

Zero-shot results on various benchmarks:

Task	**Metric**	**Fine-tuned SOTA**	**GPT-2 Zero-shot**
Translation (En->Fr)	BLEU	45.6	11.5
Summarization	ROUGE	40.2	29.3

Text Generation Quality

GPT-2 Generated Text Sample

Prompt: "In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains."

GPT-2 continues:

"Even more surprising to the researchers was the fact that the unicorns spoke perfect English. The scientist named the population, after their distinctive horn, Ovid's Unicorn. These four-horned, silver-white unicorns were previously unknown to science..."

Observations:

- Coherent and fluent
- Maintains context and style

GPT-3: The 175B Parameter Model

Discussion

What happens when we scale up another 100x?

Model size comparison:

Model	Parameters	Relative Size
GPT-1	117M	1x
GPT-2	1.5B	13x
GPT-3	175B	1,500x

GPT-3 Model Specifications

Component	Value
Layers	96
Hidden size (d_model)	12,288
Attention heads	96
Context window	2048 tokens
Training tokens	300 billion

GPT-3 Training Data

Training corpus composition:

Dataset	Tokens	Weight in Training
Common Crawl (filtered)	410B	60%
WebText2	19B	22%
Books1	12B	8%

Few-Shot Learning

GPT-3's key capability: In-context learning

Learning Paradigms Compared

1. Zero-shot: Task description only

```
Translate to French: I love AI →
```

2. One-shot: One example

```
sea otter → loutre de mer
I love AI →
```

3. Few-shot: Multiple examples (typically 10-100)

```
dog → chien
```

Few-Shot: Worked Example

Sentiment Classification with 3 examples:

```
prompt = """
```

Why this works:

- Model recognizes the pattern from examples
- Applies same pattern to new input
- No weight updates needed!

In-Context Learning: How It Works

The mechanism behind few-shot learning:

```
Prompt structure:
```

Key observations:

- Model recognizes pattern in examples
- Continues the pattern for new input
- No weight updates - pure inference!
- "Learning" happens at inference time

Important

This is NOT the same as training! The model weights don't

GPT-3's Emergent Abilities

Discussion

What can GPT-3 do that smaller models can't?

Emergent capabilities:

- **Arithmetic**: 2-3 digit addition/subtraction
- **Reasoning**: Simple logical deduction
- **Code generation**: Write simple programs
- **Knowledge synthesis**: Combine facts
- **Style transfer**: Mimic writing styles
- **Task composition**: Multi-step procedures

Emergent Abilities: Concrete Examples

Arithmetic (emerges around 10B parameters):

```
Q: What is 47 + 58?
```

Code Generation:

```
# Write a function to check if a number is prime
def is_prime(n):
```

Multi-step Reasoning:

```
Q: If I have 3 apples and give 2 to my friend,
```

The Scaling Laws Hypothesis

Kaplan et al. (2020) - "Scaling Laws for Neural Language Models"

Model performance scales as a **power law** with:

- Model size (parameters)
- Dataset size (tokens)
- Compute (FLOPs)

Key findings:

1. Performance depends strongly on scale
2. Very weak dependence on model shape (depth vs width)
3. Smooth, predictable improvements

The Scaling Law Formula

Loss as a function of scale:

$$L(N) = \left(\frac{N_c}{N}\right)^{\alpha_N}$$

where:

- L = Cross-entropy loss
- N = Number of parameters
- N_c = Scaling constant
- $\alpha_N \approx 0.076$ (empirically determined)

Scaling Laws: Visual Understanding

How loss decreases with scale:

```
Loss
 ^
 |
3.5| *
 | *
3.0| *
```

Key observation: Returns diminish but never stop - every 10x increase helps!

Implications of Scaling Laws

Good news:

- Predictable improvements
- Clear path to better models
- Can plan compute budgets
- Smooth progress curve

Challenges:

- Diminishing returns
- Exponential cost increase
- Hardware limitations
- Environmental impact

The Cost of Scaling

To halve the loss:

Chinchilla Scaling Laws

Hoffmann et al. (2022)

Previous models were **over-parameterized and under-trained!**

Key insight:

- For a given compute budget, should balance model size and data
- **Optimal ratio**: ~20 tokens per parameter

Comparison:

			Tokens/Par

The Path to ChatGPT

Evolution from GPT-3 to ChatGPT:

Stage	Model	Key Innovation
1	GPT-3	Predict next token
2	InstructGPT	Follow instructions
		Helpful &

Instruction Tuning

What is Instruction Tuning?

Fine-tune the model on (instruction, response) pairs to make it better at following user commands.

Training examples:

```
Instruction: "Explain quantum computing to a 5-year-old"
Response: "Imagine you have a magic coin that can be heads
 AND tails at the same time until you look at it..."

Instruction: "Write a Python function to sort a list"
Response: "def sort_list(items):
```

Before vs After Instruction Tuning

Before (GPT-3):

```
User: "Write a haiku about AI"

GPT-3: "Write a haiku about
```

After (InstructGPT):

```
User: "Write a haiku about AI"

InstructGPT: "Silicon neurons
```

RLHF: Reinforcement Learning from Human Feedback

The 3-step RLHF process:

```
Step 1: Supervised Fine-tuning (SFT)
- Train on human-written examples of good responses
- Model learns basic instruction-following

Step 2: Reward Model Training
- Generate multiple responses to same prompt
- Humans rank responses from best to worst
- Train a model to predict human preferences

Step 3: RL Optimization (PPO)
- Generate responses with policy model
- Score with reward model
```


continued...

RLHF: Reinforcement Learning from Human Feedback

```
- Update policy to maximize reward
```

Week 7

...continued

RLHF: Worked Example

Training the reward model:

```
Prompt: "How do I make a cake?"

Response A (Rating: 4/5):
"Here's a simple recipe: Preheat oven to 350F.
Mix 2 cups flour, 1.5 cups sugar..."

Response B (Rating: 2/5):
"Cake is a type of dessert that originated in
ancient civilizations..."
```

ChatGPT's Impact

Launched: November 30, 2022

Growth:

- 1 million users in 5 days
- 100 million users in 2 months
- Fastest-growing consumer application ever

Why so successful?

- Easy to use (conversational interface)

The Modern LLM Landscape

Post-GPT-3 developments (2020-2024):

Year	Milestone
2021	Anthropic founded (Claude)
2022	ChatGPT launched
2023	GPT-4 (multimodal, improved reasoning)

Open Source LLMs

Discussion

Should powerful AI models be open or closed?

Closed (GPT-4, Claude):

- Better safety control
- Monetization easier
- Protect IP
- No transparency

Open (Llama, Mistral):

- Transparency
- Community innovation
- Full control
- No API costs

Practical Prompting Techniques

Effective prompting strategies for modern LLMs:

1. Zero-shot with clear instructions:

```
Classify the following text as spam or not spam.
Only respond with "spam" or "not spam".
```

2. Few-shot with examples:

```
Text: "Meeting at 3pm tomorrow" → not spam
Text: "URGENT: Send money now!" → spam
```

Chain-of-Thought Prompting

For complex reasoning tasks:

```
Q: Roger has 5 tennis balls. He buys 2 more cans of
 tennis balls. Each can has 3 balls. How many
 tennis balls does he have now?

Let's think step by step:
1. Roger starts with 5 tennis balls
2. He buys 2 cans of tennis balls
```

Key insight

Adding "Let's think step by step" dramatically improves reasoning accuracy!

- Multi-step logic
- Mathematical proofs

3. **Knowledge grounding**

- Knowledge cutoff date
- Can't access real-time info

4. **Personalization**

- No fine-tuning needed for many tasks
- In-context learning is powerful

3. **Scaling laws provide predictability**

- But diminishing returns and compute costs are real
- Chinchilla scaling: balance model size and data

4. **RLHF changed everything**

Readings

Required Readings

1. **Radford et al. (2019)** - "Language Models are Unsupervised Multitask Learners" (GPT-2) [PDF]
2. **Brown et al. (2020)** - "Language Models are Few-Shot Learners" (GPT-3) [ArXiv]
3. **Kaplan et al. (2020)** - "Scaling Laws for Neural Language Models" [ArXiv]

Recommended Readings

- **Wei et al. (2022)** - "Emergent Abilities of Large Language Models" [ArXiv]

Next Lecture Preview

Lecture 23: Implementing GPT from Scratch

- Building a mini-GPT in PyTorch
- Tokenization with BPE
- Training loop and optimization
- Sampling strategies (greedy, top-k, nucleus)
- Hands-on coding session

Questions?

Week 7